CS 4720 HW7

1a) Correct
c) Correct
e) Correct
h) Correct
j) Correct
k) Correct

2) Let us take a look at this equation($\alpha \rightarrow (\beta \vee \gamma)$ first over here.

We know that the $p \rightarrow q = \neg p \vee q$, so similarly we can say that $\alpha \rightarrow (\beta \vee \gamma) = \neg \alpha \vee \beta \vee \gamma$

Now, if we look at the equation $(\alpha \rightarrow \gamma) \vee (\alpha \rightarrow \beta)$, then we can say that it comes out to be $\neg\alpha \vee \gamma \vee \beta$. Since both the equations point out to the same outcome so we can say that $(\alpha \rightarrow (\beta \vee \gamma)) \rightarrow ((\alpha \rightarrow \gamma) \vee (\alpha \rightarrow \beta))$, which means that it is a tautology (since it indicates that $P \rightarrow P$), which means that it is true for all values of α,β and γ.

Hence, if $(\alpha \rightarrow (\beta \vee \gamma))$ then $\alpha \rightarrow \beta$ or $\alpha \rightarrow \gamma$ or both.

3a) Let us determine whether this statement is valid or invalid, satisfiable or unsatisfiable, with the help of the following table. Over Fo represents Food, P represents Party, and D represents Drinks.

Fo	D	P	Fo → P	D → P	L = (Fo→P)V(D→P)	Fo ∧ D	R=(Fo ∧P) → P	L→R
T	T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F	T
T	F	T	T	T	T	F	T	T
T	F	F	F	T	T	F	T	T
F	T	T	T	T	T	F	T	T
F	T	F	T	F	T	F	T	T
F	F	T	T	T	T	F	T	T

F	F	F	T	T	T	F	T	T

The last column shows all true values, which means that it is a tautology. T also means that the statement is valid and satisfiable since there is at least one truth assignment over here.

b) Let us first take a look at the LHS part over here:

$(Fo \rightarrow P) \vee (D \rightarrow P) = (\neg F \vee P) \vee (\neg D \vee P) = \neg F \vee \neg D \vee P$

Now, let us take a look at the RHS: $(Fo \wedge D) \rightarrow P$

$(Fo \wedge D) \rightarrow P = \neg(F \wedge D) \vee P = (\neg F \vee \neg D) \vee P$

Therefore, LHS = RHS after expanding the equations on both sides, which means that it is a tautology and hence it is a valid and satisfiable statement.

c) Let us first negate the whole implication and put it in CNF, so we can represent it like this:

$\neg(((F \rightarrow P) \vee (D \rightarrow P)) \rightarrow ((F \wedge D) \rightarrow P)) = ((F \Rightarrow P) \vee (D \rightarrow P)) \wedge \neg((F \wedge D) \rightarrow P)$

Now, let us rewrite the following parts:

1)$F \rightarrow P = \neg F \vee P$, $D \rightarrow P = \neg D \vee P$, which means that it can be represented as $(\neg F \vee P) \vee (\neg D \vee P) = \neg F \vee \neg D \vee P$

2) Next, let us rewrite $(F \wedge D) \rightarrow P = \neg(F \wedge D) \vee P = \neg F \vee \neg D \vee P$, so it means that $\neg((F \wedge D) \rightarrow P) = F \wedge D \wedge \neg P$.

So, now we have the following clause set, which we can use for resolution:

$\{C1: \neg F \vee \neg D \vee P,\ C2: F,\ C3: D,\ C4: \neg P\}$

Let us now resolve C1 with C2 on F, so now we can represent it as $\neg F \vee \neg D \vee P$ and $F \rightarrow C5: \neg D \vee P$.

Next, let us resolve C5 with C3 on D:

$\neg D \vee P$ and $D \rightarrow C6: P$

Lastly, let us resolve C6 with C4 on P:

$P \wedge \neg P \rightarrow \perp$ (it means that it is an empty clause)

Hence, we can say that we have derived a contradiction, so the CNF of the original sentence is not satisfiable, therefore the original sentence is valid.

4c) $\forall p[Occupation(p,Surgeon) \rightarrow Occupation(p,Doctor)]$
e) $\exists p[Boss(p,Emily) \land Occupation(p1,Lawyer))$
f) $\exists p[Occupation(p,Lawyer) \land \forall c[Customer(c,p) \rightarrow Occupation(c,Doctor)]])$

5a) This statement is wrong because it means that there exists some y such that if it is a Paris apartment then x is a cheaper, which could be true even if y is not even an apartment in Paris since its making an implication right now where even if the left side of the arrow is false, it will return a true since the value to its right is true. We can correct and show it in the following way:

$\forall x[(Apt(x) \land In(x,London)) \Rightarrow \exists y(Apt(y) \land In(y,Paris) \land Rent(x)<Rent(y)))]$

This statement makes sure that the rents of both x and y are compared only if they exist in London and Paris.

b) This statement is incorrect because it never tells us whether the rent for apartment x is less than 1000, which is the important information the question is looking for. We can correct and display it in the following manner:

$\exists x[(Apt(x) \land In(x,Paris) \land Rent(x)<Dollars(1000)) \land \forall y((Apt(y) \land In(y,Paris) \land Rent(y) <dollars(1000)) \rightarrow y=x)$

Over here, we are specifying that we want the value of x to be less than 1000, while making sure that it is the only apartment that has a rent of less than 1000 dollars.

c) The original statement is wrong because we are assuming that everything in the world is an apartment, and if every y is an apartment in London and it is cheaper than x, then it lies in Moscow and that’s not what we are looking for in this question, instead we are looking for something like for every apartment x, if its rent is higher than all London apartments y, then it must be in Moscow, which can be represented as follows:

$\forall x ((Apt(x) \land \forall y((Apt(y) \land In(y,London)) \Rightarrow Rent(x)>Rent(y))) \Rightarrow In(x,Moscow))$

6c) Since either one of them has written the book, we can represent the first-order logic as follows:

$Wrote(Lenon, ADayInTheLife) \lor Wrote(McCartney,ADayInTheLife) \land \neg(Wrote(Lemon,ADayInTheLife) \land Wrote(McCartney,ADayInTheLife)$

f) $\forall s(Sings(McCartney,s,Revolver) \rightarrow Wrote(McCartney,s))$

h) Over here, we can say that there exists some album such as where the songs have been stored, and there also exists some singer who has sung those songs.

$\forall s(Wrote(Gershwin,s) \rightarrow \exists a \exists p\ Sings(p,s,a)$

i) We can represent the first-order logic for this statement as follows:

$\exists a \forall s(Wrote(Joe,s) \rightarrow \exists p\ Sings(p,s,a))$